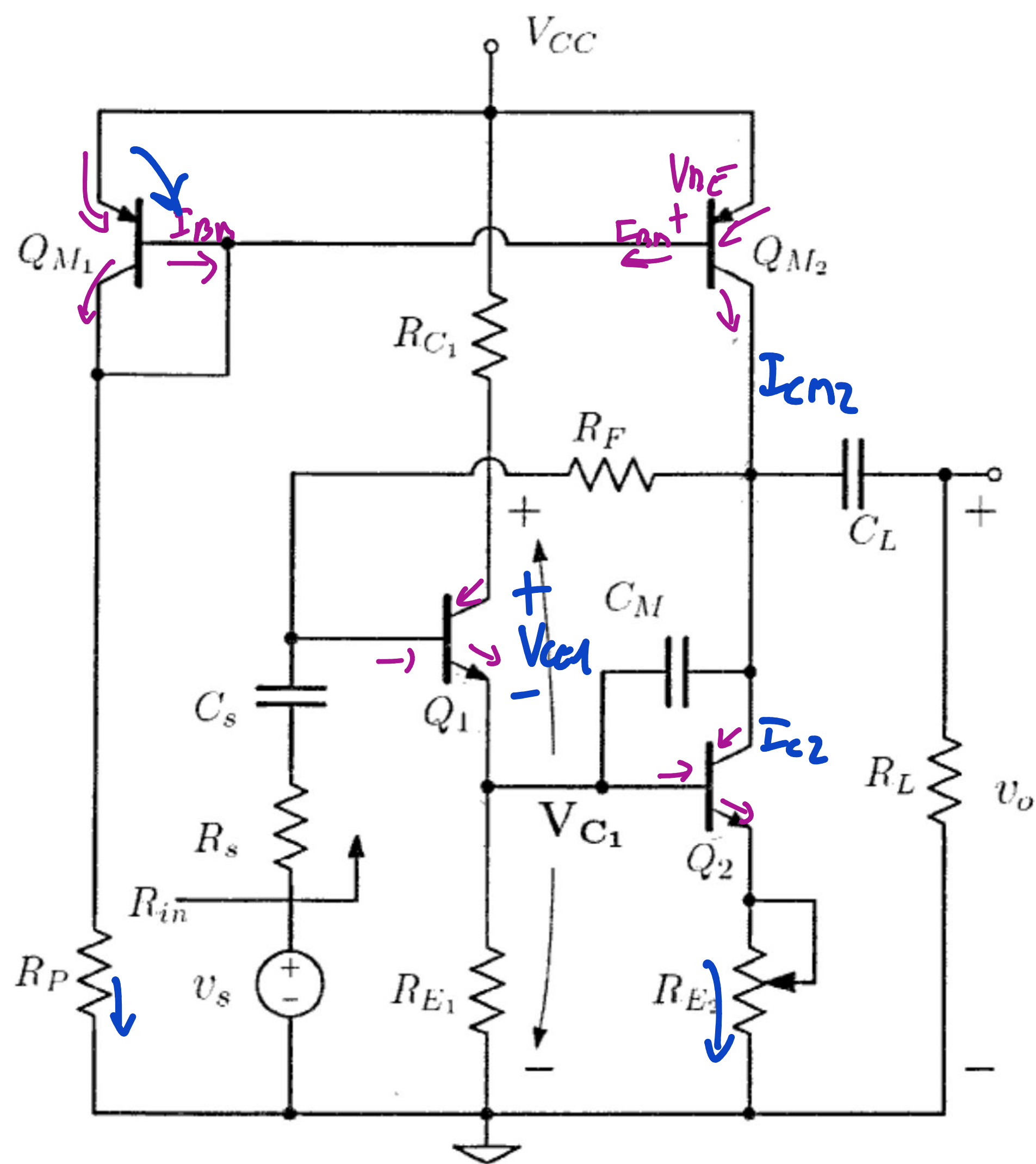


Ex 1 3 coll 2022



Consider the circuit in Fig. 1. The circuit parameters, at $T = 25^\circ\text{C}$, are the following:

$V_{CC} = 10\text{ V}$; $R_P = 4.7\text{ k}\Omega$; $R_{C1} = 1\text{ k}\Omega$; $R_{E1} = 1\text{ k}\Omega$; $R_F = 200\text{ k}\Omega$; $R_S = 10\text{ k}\Omega$; $R_L = 18\text{ k}\Omega$.
 $C_S = 1\text{ }\mu\text{F}$; $C_L = 1\text{ }\mu\text{F}$; $C_M = 50\text{ pF}$.

$Q_{1,2}$ (npn BJT): $V_{BE1,2} = 0.65\text{ V}$; $\beta_{F1,2} = \beta_{01,2} = 200$; $r_{01,2} = +\infty$;

$Q_{M1,2}$ (pnp BJT): $V_{BEM1,2} = -0.7\text{ V}$; $\beta_{FM1,2} = \beta_{0M1,2} = 150$; $r_{0M1,2} = +\infty$;

$V_T = 26\text{ mV}$.

Considering all capacitors to be equivalent to *open circuits*, determine:

1. the value of R_{E2} that makes the bias level of the collector of Q_1 , V_{C1} , equal to 8 V ;
2. the operating points (V_{CE} , I_C) of Q_1 and Q_2 .

Assuming that, at the frequencies of interest (i.e. at mid-band), C_S and C_L are equivalent to *short circuits*, while C_M is equivalent to an *open circuit*, determine:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance indicated in the figure.

Assuming only C_M is effective (while C_S , C_L are shorted) and neglecting all C_π 's and C_μ 's, determine:

5. the high frequency bandwidth limit f_H of the amplifier.

1) operating point, imposing $V_{C1} = 8\text{ V}$

$$I_{RP} = \frac{V_{CC} + V_{BE}}{R_P} = 1.978\text{ mA}$$

$$I_{RP} = I_{CM1} + I_{B2} + I_{B1} = \beta_{FM1} I_{B1} + 2I_{B1}$$

$$I_{B1} = \frac{I_{RP}}{\beta_{FM1} + 2} = 0.13\text{ mA} \quad I_{CM1,2} = \beta_F I_{B1} = 1.952\text{ mA}$$

$$I_{C1} = \frac{V_{CC} - V_{C1}}{R_{C1}} = 2\text{ mA}$$

$$I_{B1} = \frac{I_{C1}}{\beta_{FM1}} = 0.01\text{ mA} = I_{RF}$$

$$I_{C2} = I_{CM2} - I_{RF} = \beta_{FM2} I_{B2} - I_{RF} = 1.94\text{ mA} \quad I_{E2} = 1.951\text{ mA}$$

$$I_{B2} = \frac{I_{C2}}{\beta_F} = 9.7\text{ nA}$$

$$V_{CE1} = V_{C1} - R_{E1}(I_{E1} - I_{E2}) = 6.05\text{ V}$$

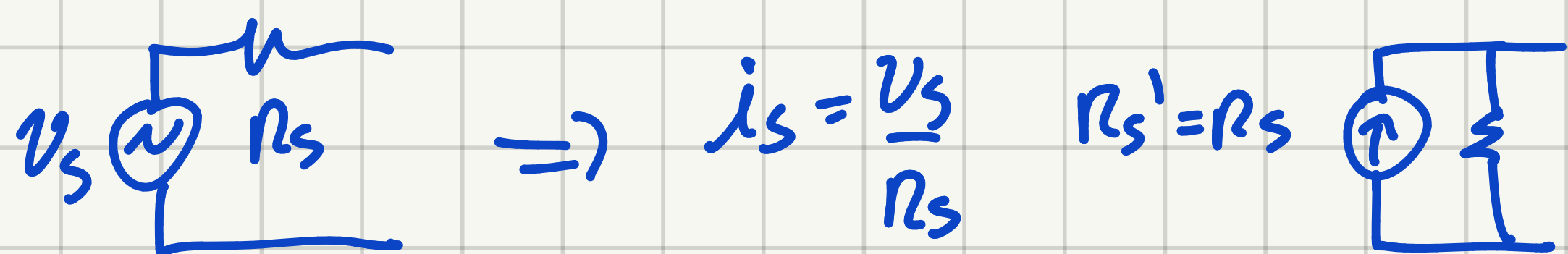
$$V_{C1} - V_{CE1} - R_{E2} I_{E2} = 0$$

$$R_{E2} = \frac{V_{C1} - V_{CE1} - V_{BE2}}{I_{E2}} = 666\text{ }\Omega$$

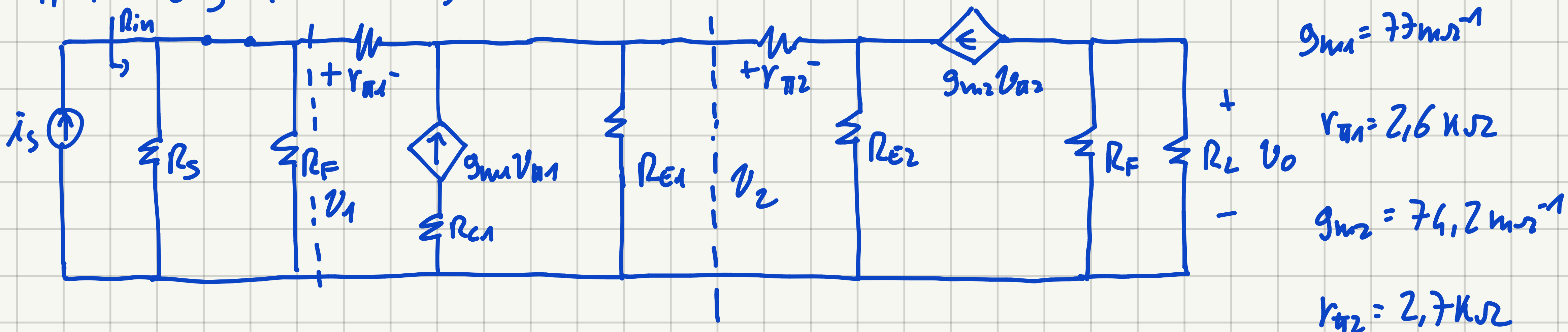
$$V_{CE2} = V_{BE2} + V_{BE1} + R_F I_{B1} = 3,3 \text{ V}$$

2) Voltage gain $A_v = v_o/v_s$

it is a voltage sampling current mixing $\rightarrow$ transresistance $\rightarrow$ norton equivalent of v_s



apply loading effect killing feedback in output



$$A_{OL} = \frac{v_o}{v_2} \cdot \frac{v_2}{v_1} \cdot \frac{v_1}{i_s} = \frac{v_o}{i_s}$$

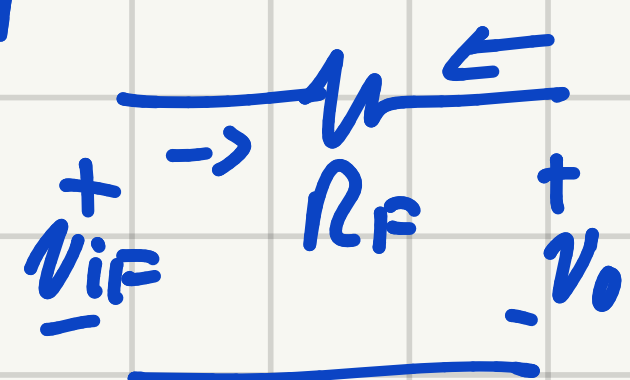
$$R_{in1} = r_{\pi1} + (1 + \beta_0)(R_{E1} \parallel R_{in2}) = 202,1 \text{ k}\Omega$$

$$R_{in2} = r_{\pi2} + R_{E2}(1 + \beta_0) = 136,6 \text{ k}\Omega$$

$$= - \frac{\beta_0 (R_F \parallel R_L)}{r_{\pi2} + R_{E2}(1 + \beta_0)} \cdot \frac{(1 + \beta_0)(R_{E1} \parallel R_{in2})}{r_{\pi1} + (1 + \beta_0)(R_{E1} \parallel R_{in2})} \cdot \underbrace{(R_s \parallel R_F \parallel R_{in1})}_{9,09 \text{ k}\Omega} = -217,1 \text{ k}\Omega$$

-24,2 0,987

β network $i_F = f(v_o)$



$$i_F = \beta v_o + v_{iF}/R_F$$

$$\beta = i_F/v_o|_{v_{iF}=0} = -1/R_F = -5 \cdot 10^{-3} \text{ mS}^{-1}$$

$$A_F = \frac{A_{OL}}{1 + \beta A_{OL}} = -104,6 \text{ k}\Omega$$

$$1 + \beta A_{OL} = 2,08$$

$$A_v = \frac{v_o}{v_s} = \frac{v_o}{i_s} \cdot \frac{i_s}{v_s} = A_F \cdot \frac{1}{R_s} = -104,6$$

4) R_{in} of the figure. we applied norton

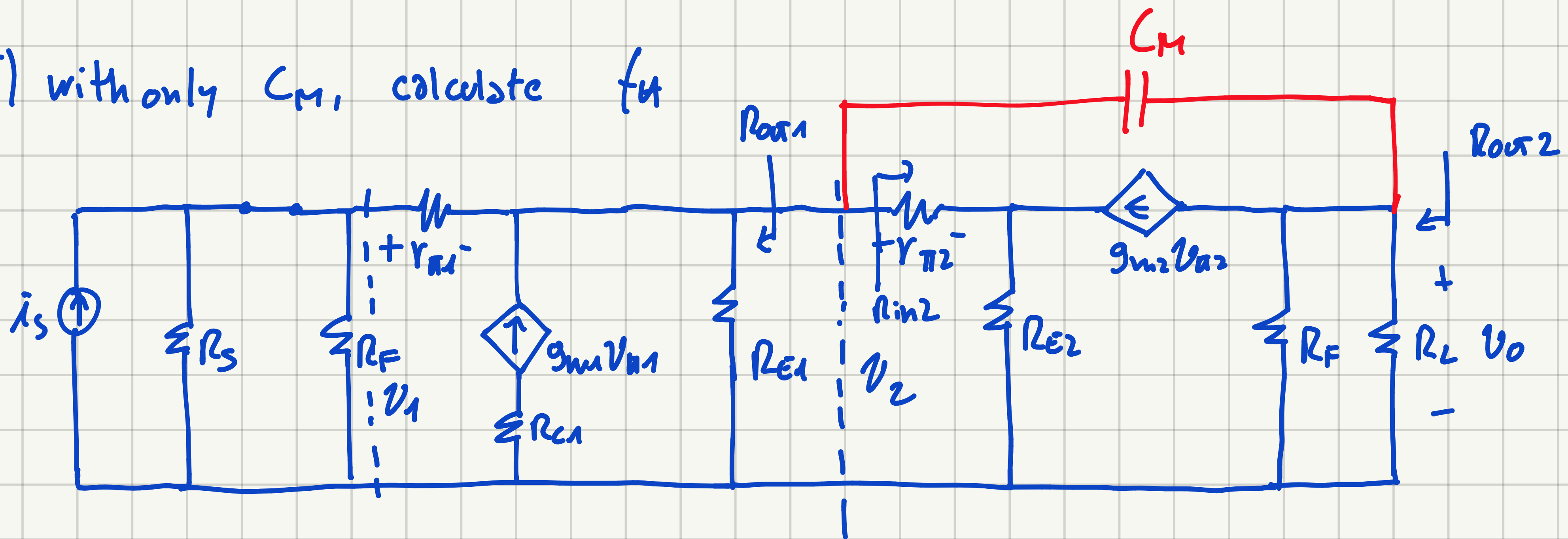
$$R_{in}^{OL} = R_S // R_F // R_{in1} = 9,09 \text{ k}\Omega$$

$$R_{in}^F = \frac{R_{in}^{OL}}{1 + \beta A_{OL}} = 4,35 \text{ k}\Omega$$

$$R_{in}^F = R_S // (R_F // R_{in1})$$

$$R_{in} = \left(\frac{1}{\frac{1}{R_{in}^F} - \frac{1}{R_S}} \right) + R_S = 17,7 \text{ k}\Omega$$

5) with only C_M , calculate f_H



$$R_{out1} = \frac{r_{\pi1} + (R_S // R_F)}{1 + \beta_0} // R_{E1} = 57 \Omega$$

$$R_{out2} = R_F // R_L = 16,5 \text{ k}\Omega$$

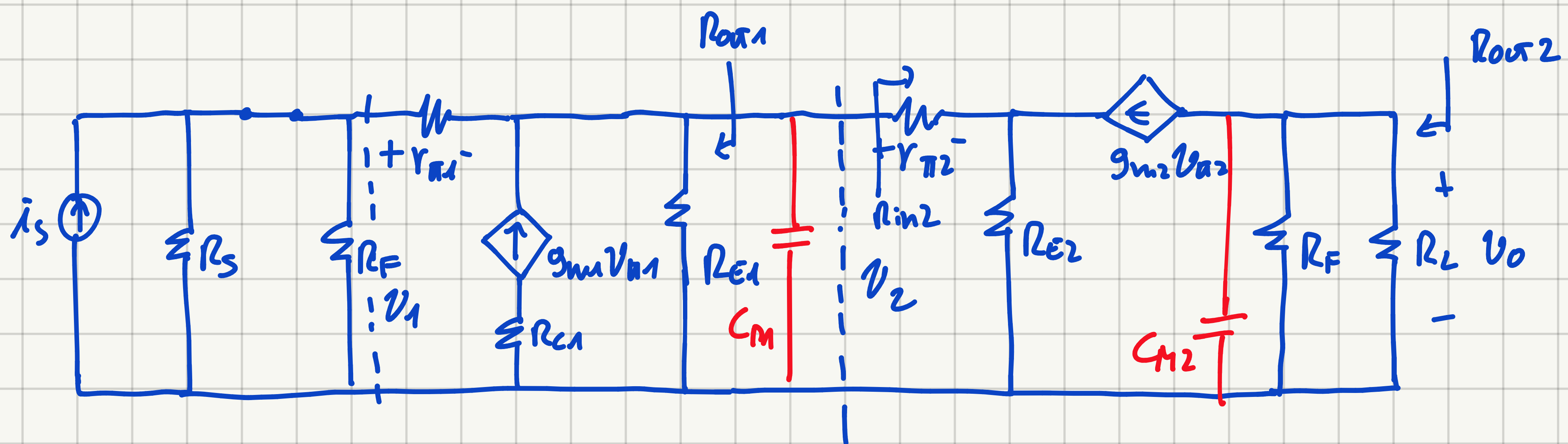
Miller theorem to split C_M

$$\begin{cases} Y_1 = (1 - G)Y \\ Y_2 = (1 - 1/G)Y \end{cases}$$

$$G = -\frac{\beta_0 R_{out2}}{r_{\pi1} + R_{E2} (1 + \beta_0)} = -24,1$$

$$C_{M1} = C_M (1 - G) = 1,255 \text{ nF} \quad \text{Miller multiplier}$$

$$C_{M2} = C_M (1 - 1/G) = 52 \text{ pF}$$



$$f_H = \frac{1}{2\pi} \left(\frac{1}{C_{M1} R_{M1}^o + C_{M2} R_{M2}^o} \right)$$

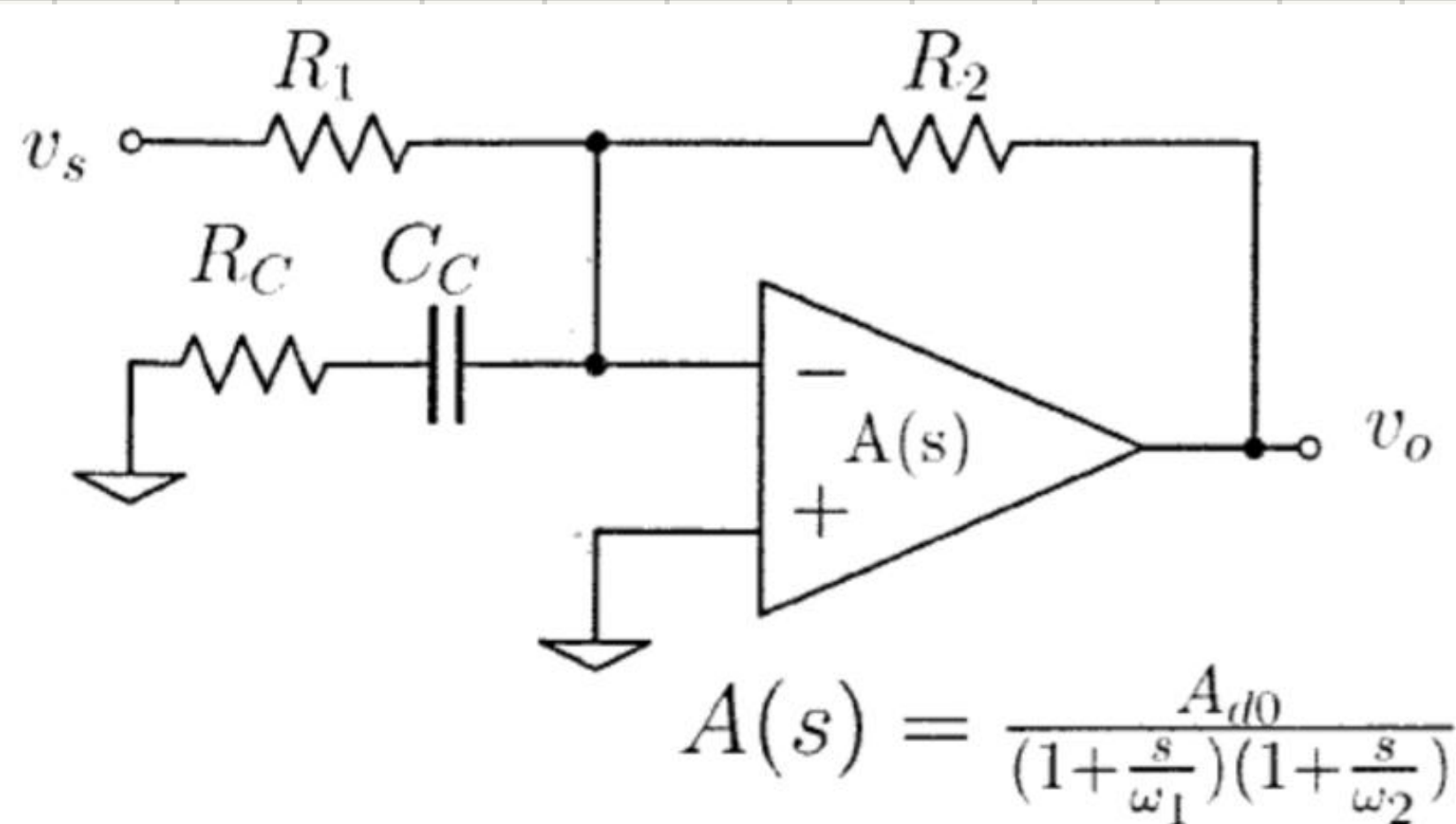
$$R_{M1}^o = R_{out1} \parallel R_{in2} = 56 \Omega$$

$$R_{M2}^o = R_{out2} = 16,5 \text{ k}\Omega$$

$$\rightarrow f_H^{ol} = 171,5 \text{ kHz}, \quad f_H^F = f_H^{ol} (1 + \beta A_{ol}) = 357 \text{ kHz}$$

Ex 2 3 com 2022

18:30



Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 56 \text{ k}\Omega$; $R_2 = 120 \text{ k}\Omega$; $R_C = 470 \Omega$; $C_C = 100 \text{ nF}$;

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$; $\omega_1 = 1 \text{ krad/s}$; $\omega_2 = 500 \text{ krad/s}$.

Considering, initially, C_C to be open circuited, determine:

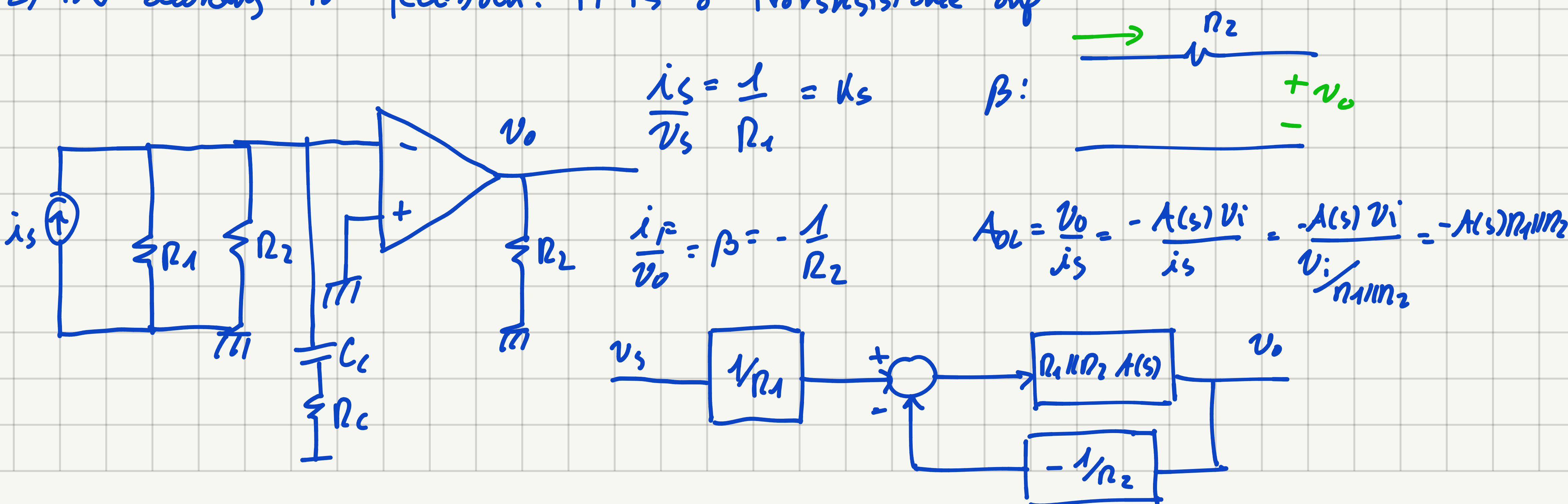
1. the expression of the *ideal* amplifier voltage gain $A_v(s) = v_o/v_s$;
2. the equivalent *block diagram* of the amplifier, according to *feedback theory*;
3. the amplifier *bandwidth* and *phase margin*.

Then, applying the noise gain compensation network R_C - C_C , find:

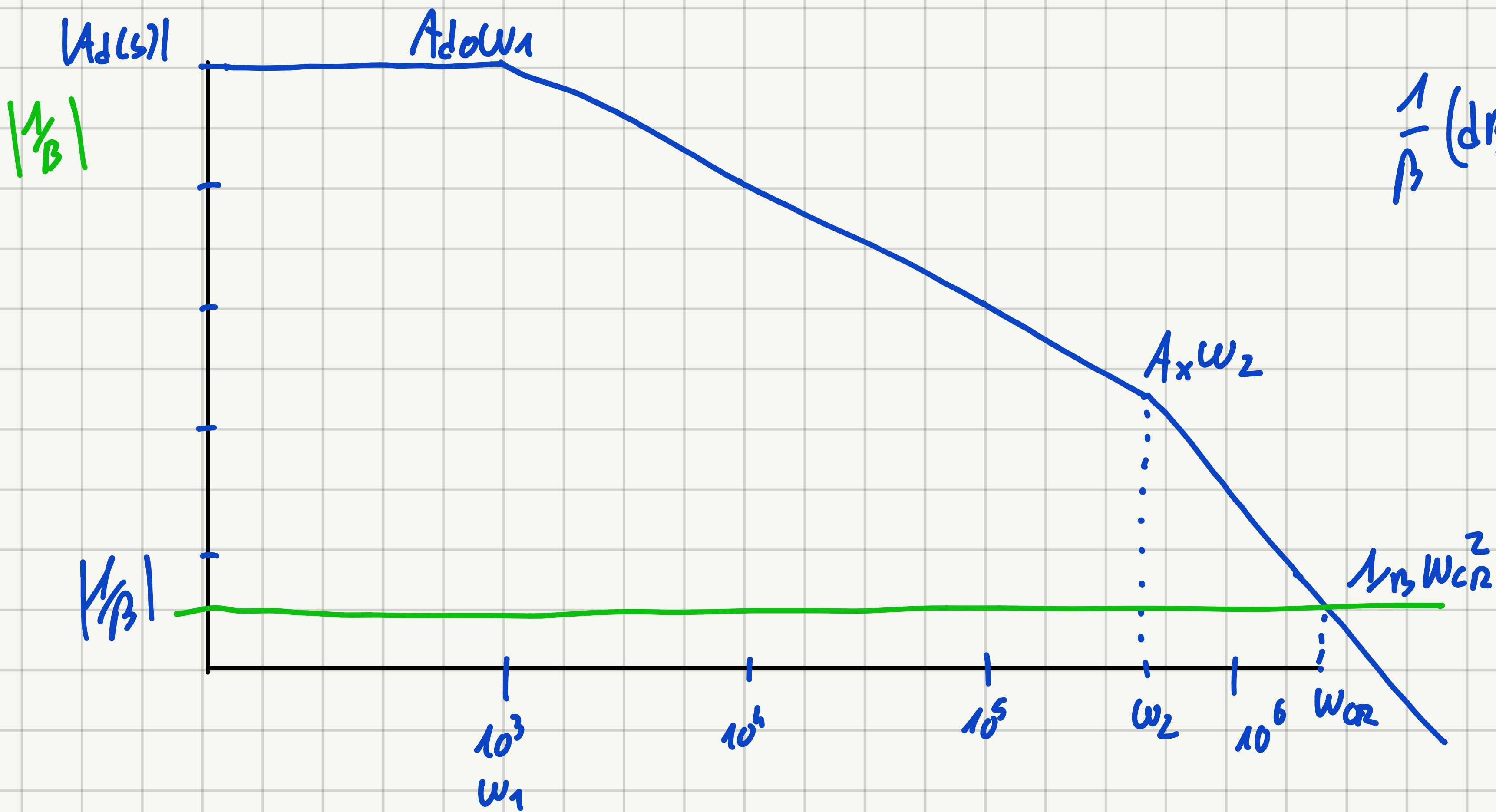
4. the *bandwidth* and the *phase margin* of the compensated amplifier.

$$1) A_v(s) = -\frac{R_2}{R_1} = -2,143 \text{ V/V}$$

2) BD according to feedback. it is a transresistance amp



3) Bandwidth and phase margin (without compensation) $\beta = -\frac{1}{R_2} \cdot R_1 // R_2 = \frac{R_1}{R_1 + R_2}$



$$\frac{1}{\beta} \text{ (dB)} = 10 \text{ dB}$$

from
DCI or
by tuning
at $R_1 // R_2$
from $A_d(s)$

$$A_{d0} \omega_1 = A_{\infty} \omega_2$$

$$A_{\infty} \omega_2^2 = \frac{1}{\beta} \omega_{cn}^2 \quad \omega_{cn} = \sqrt{A_{\infty} \beta \omega_2^2} = \sqrt{A_{d0} \omega_1 \omega_2 \beta} = 3,98 \cdot 10^6 \text{ rad/s}$$

$$\omega_{cn} \rightarrow \beta$$

$$\varphi_m = 180 - \arctan \frac{\omega_{cn}}{\omega_1} - \arctan \frac{\omega_{cn}}{\omega_2} = 7,16^\circ$$

4) Now apply noise gain comp. calculate BW and φ_m

$$R_c = 470 \Omega \quad C_c = 100 \text{ nF} \quad \text{we add a zero and a pole}$$

C_c small $\rightarrow$ LF untouched

resistance seen from the cap

$$\text{pole is at } \omega_p = \frac{1}{C_c (R_c + R_1 // R_2)} = 258,72 \text{ rad/s}$$

$$\beta_{\infty} = \frac{R_1 // R_c}{R_2 + (R_1 // R_c)} = 3,869 \cdot 10^{-3} \rightarrow \frac{1}{\beta_{\infty}} = 258,46 = 48 \text{ dB}$$

on a +20 dB/dec line the ratio A/ω is constant so we can easily find position of the zero

$$\omega_z = \omega_p \frac{\beta_0}{\beta_p} = 21,28 \cdot 10^3 \text{ rad/s}$$



$$\varphi_{mc} = 180 - \arctan\left(\frac{w_{mc}}{w_1}\right) - \arctan\left(\frac{w_{mc}}{w_2}\right) - \arctan\left(\frac{w_{mc}}{w_p}\right) + \arctan\left(\frac{w_{mc}}{w_z}\right) = 49,3^\circ$$